

WRDS Corporate Bond Relative Value Assignment

In this project, I use the supplied WRDS enhanced end-of-month files to study relative value across the bonds of six large corporate issuers.

My main steps are:

1. Choose issuers with enough bonds to cover a useful duration range.
2. Keep seniority and rating consistent within each issuer-month.
3. Plot Treasury spread (T_Spread) against duration.
4. Fit a weighted issuer spread curve and classify bonds as rich or cheap from the fitting residual.
5. Build monthly long-cheap/short-rich portfolios with close to zero net DV01.
6. Test three portfolio constructions instead of relying on one arbitrary choice.
7. Compare next-month return, dollar PnL, turnover, trading-cost sensitivity, and outlier characteristics.

The final portfolio is the strongest of the three tests on this sample. I report all three results and discuss the resulting selection bias rather than presenting the winner as an out-of-sample result.

Ideas from Class

Several choices in the analysis come directly from topics Professor Boroditsky discussed in class:

- **Duration and DV01.** Using $dP/P = -D dy$ and $D = 10,000 * DV01 / P$, the assignment's condition $\text{sum}(\text{position} * \text{price} * \text{duration}) = 0$ is also a first-order hedge against a parallel one-basis-point rate move.
- **Relative value.** A bond is rich or cheap relative to an issuer curve. I use actual spread minus fitted spread as the signal.
- **Curve construction.** The course covered Nelson-Siegel and weighted fitting. For this homework I keep the model linear and use the three permitted factors: log-duration, duration, and coupon.
- **Price reliability.** Professor Boroditsky noted that larger and more actively traded issues usually provide better calibration points. I use both TRACE dollar volume and amount outstanding in the regression weights.
- **Accrued interest.** The return fallback adds coupon accrual using the actual number of days divided by 360.
- **Trading frictions.** Corporate bonds are not frictionless. I report turnover and results after several assumed trading-cost levels.

Course reference: Misha Boroditsky, *MTH9897 Quantitative Investment Framework*, Lectures 1-2, especially slides 12-15, 20-21, and 51-67.

Parameters

I selected AAPL, AMZN, BA, CAT, DIS, and T because each issuer has enough observations to form a monthly curve across a meaningful duration range. The common sample runs from August 2022 through March 2025.

Professor Boroditsky's high-quality calibration example uses at least one year to maturity and roughly \$200 million outstanding. The WRDS amount fields in these files are reported in thousands, so `200\_000` corresponds to about \$200 million.

Input files:

data\WRDS_Data-Systematic_Trading-Fall2026-1.csv (10.0 MB)
data\WRDS_Data-Systematic_Trading-Fall2026-2.csv (11.5 MB)

Helper Functions

The WRDS download stores several numeric fields as strings with percent signs, dollar signs, or commas. The functions below convert them to numeric values and keep the original data logic visible.

Load and Clean Data

I apply the following filters before fitting the curves:

- Convert the WRDS text fields to numeric spreads, returns, prices, and trading volumes.
- Remove defaulted and convertible observations, bonds with less than one year remaining, small issues, and implausible prices or spreads.
- Within each issuer-month, keep the most common `SECURITY_LEVEL` and `RATING_CAT`. This avoids comparing senior and subordinated debt or mixing rating groups in the same curve.
- Show the number of observations remaining after each step.

The files do not contain a complete call schedule, bid-offer spread, 144A flag, subsidiary mapping, or fixed-versus-step-up coupon flag. I do not try to infer those missing fields. They remain limitations of the analysis.

		cleaning_step		rows	share_of_initial
0		Selected issuers and dates		10689	100.0%
1		Market and high-quality calibration filters		7569	70.8%
2		Consistent seniority and rating		7559	70.7%

company_symbol	rows	bonds	first_date	last_date	median_duration	duration_min	duration_max	median_spread_bps
AAPL	1475	52	2022-08-31	2025-03-31	7.3100	0.9500	20.9900	19.0000
AMZN	908	33	2022-08-31	2025-03-31	6.5800	0.9500	20.0300	15.0000
BA	1304	51	2022-08-31	2025-03-31	7.3950	0.9200	16.4100	28.5000
CAT	957	51	2022-08-31	2025-03-31	3.4500	0.9500	18.6500	13.0000
DIS	1237	44	2022-08-31	2025-03-31	8.8200	0.9600	18.1900	20.0000
T	1678	56	2022-08-31	2025-03-31	10.6300	0.9600	17.8400	28.0000

Consistency-filter check. The first two columns should be 1 for every issuer-month.

	security_levels	rating_cats	bonds
count	192.0000	192.0000	192.0000
mean	1.0000	1.0000	39.3698
std	0.0000	0.0000	8.7785
min	1.0000	1.0000	24.0000
25%	1.0000	1.0000	31.0000
50%	1.0000	1.0000	40.0000
75%	1.0000	1.0000	47.0000
max	1.0000	1.0000	54.0000

Next-Month Return and Accrued Interest

I form positions at month-end and evaluate them using the next observation for the same bond when the gap is between 25 and 45 days.

RET_EOM is the preferred measure because it is the WRDS enhanced monthly total return. If that value is missing, the fallback adds the clean-price return and accrued coupon. Following the daily PnL convention discussed in class, I calculate the accrual as $\text{annual coupon rate} * \text{gap days} / 360$.

	next_return_source	rows
	RET_EOM	7226
	unavailable	333

Spread Curve Model

For each issuer-month, I estimate:

$$T_Spread_bps = b_0 + b_1 * \log(1 + DURATION) + b_2 * DURATION + b_3 * COUPON$$

The model is simple enough to explain but flexible enough to capture a slope and some curvature in duration. Duration also matches the hedge variable used later in the portfolio. Coupon helps absorb systematic cash-flow and price differences that might otherwise show up in the residual.

Professor Boroditsky pointed out that actively traded, large issues are usually more reliable calibration points. I therefore combine TRACE dollar volume and amount outstanding in the weighted least-squares fit. I winsorize both inputs within the issuer-month so that one large issue does not control the entire curve. Weighted RMSE and weighted R-squared provide a quick check on fit quality.

company_symbol	fit_months	avg_bonds	median_r2	median_weighted_rmse_bps	median_resid_std_bps
AAPL	32	46.1	0.744	6.3	6.3
AMZN	32	28.4	0.755	5.3	5.3
BA	32	40.8	0.376	10.5	10.5
CAT	32	29.9	0.617	7.1	7.1
DIS	32	38.7	0.535	8.4	8.4
T	32	52.4	0.386	12.8	12.8

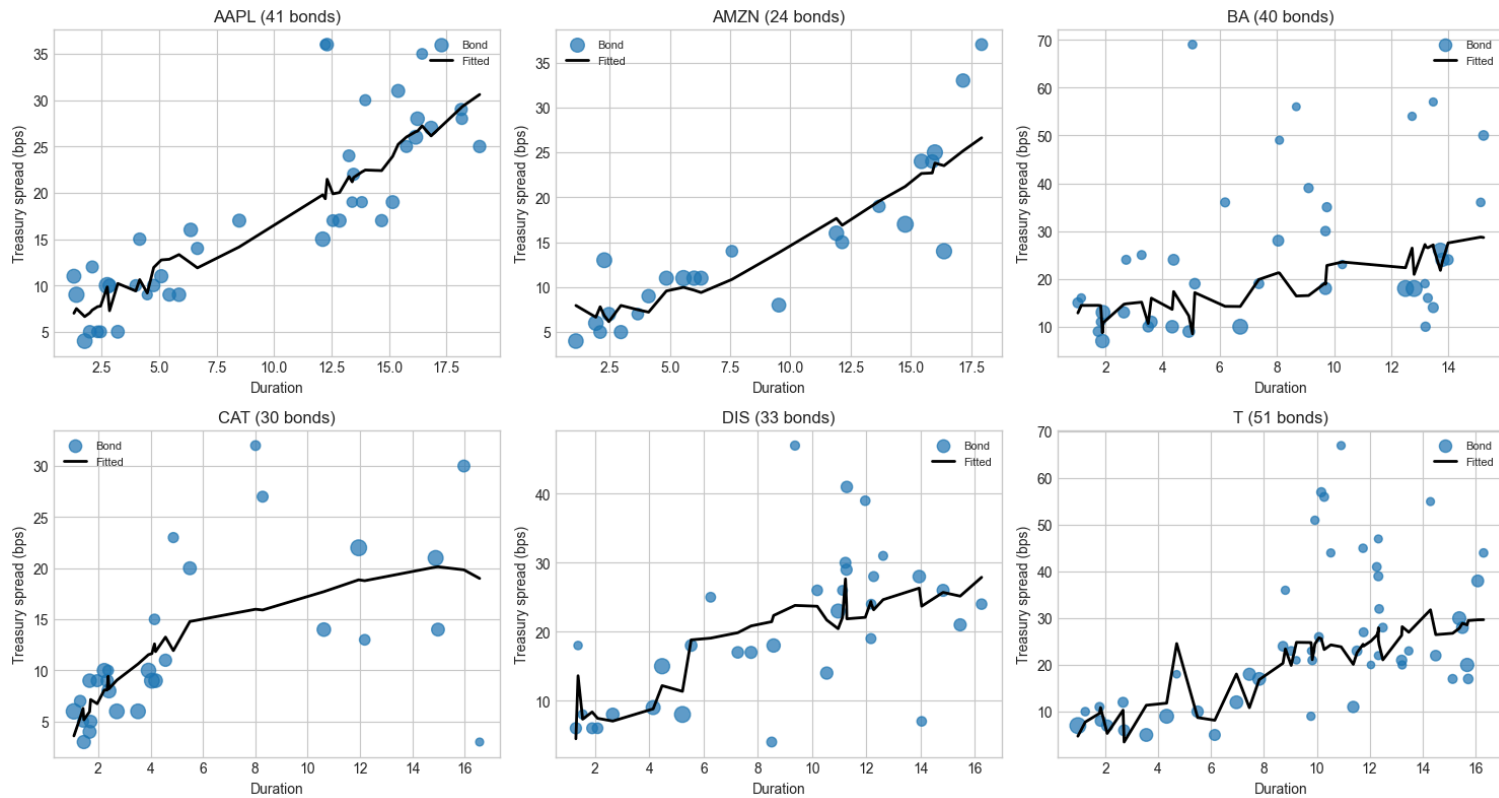
Analysis date for cross-sectional plots: 2025-03-31

	company_symbol	n_bonds	r2	resid_std_bps	b0	b_log_duration	b_duration	b_coupon
31	AAPL	41	0.7634	4.3080	8.2800	-0.9932	1.4655	-0.9608
63	AMZN	24	0.7484	4.0214	9.9781	-4.6242	1.7663	-0.5321
95	BA	40	0.3139	7.6989	16.5833	0.3325	1.1630	-1.6616
127	CAT	30	0.6063	4.3097	-1.9559	10.4093	-0.3695	-0.3732
159	DIS	33	0.6087	5.6568	-0.6256	0.5547	1.2801	1.6156
191	T	51	0.5870	7.4134	3.4247	-8.6979	2.5332	2.7872

Spread vs. Duration by Issuer

The scatter points are individual bonds on the selected analysis date. The line is the fitted curve from the cross-sectional model for that issuer-month.

Spread Curve by Issuer on 2025-03-31



Rich and Cheap Bonds

Residuals are measured as actual spread minus fitted spread:

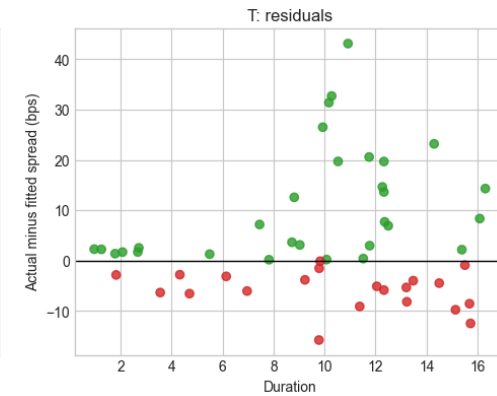
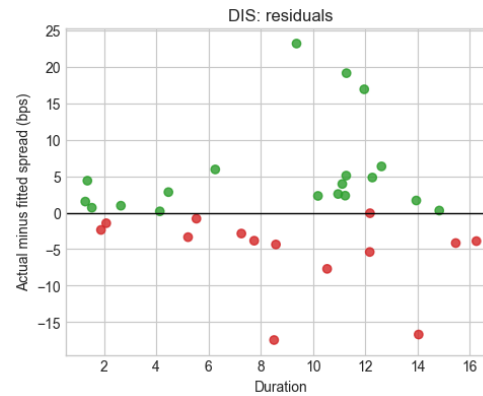
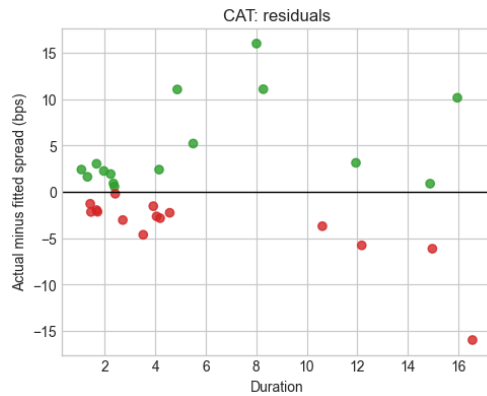
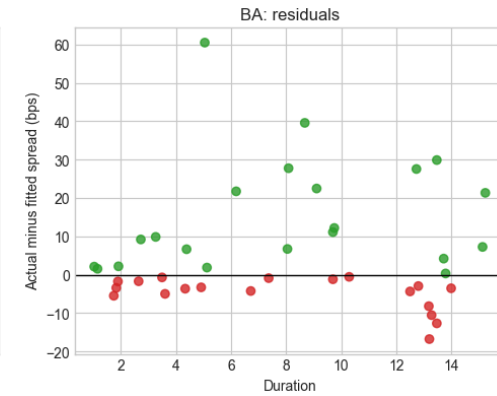
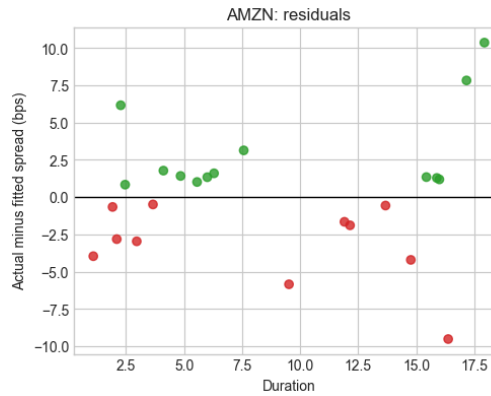
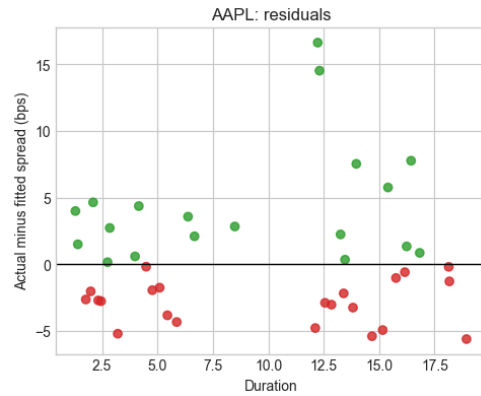
- Positive residual: the bond trades wider than the curve and is classified as cheap.
- Negative residual: the bond trades tighter than the curve and is classified as rich.

The table below shows the most extreme residuals on the analysis date.

	company_symbol	CUSIP	MATURITY	SECURITY_LEVEL	RATING_CAT	COUPON	DURATION	TMT	PRICE_EOM	T_Spread_bps	spread_fit_bps	resid_bps	resid_z	rich_heap	T_DVolume_num	AMOUNT_OUTSTANDING
1435	AAPL	037833AT7	2044-05-06	SEN	AA	4.4500	12.2300	19.3800	92.2900	36.0000	19.3625	16.6375	3.8620	cheap	20,807,341.0000	1,000,000.0000
1460	AAPL	037833EE6	2041-02-08	SEN	AA	2.3750	12.3100	16.0900	70.2400	36.0000	21.4675	14.5325	3.3734	cheap	79,582,495.0000	1,500,000.0000
1456	AAPL	037833DZ0	2050-08-20	SEN	AA	2.4000	16.4400	25.7600	59.8700	35.0000	27.2275	7.7725	1.8042	cheap	28,309,879.0000	1,250,000.0000
1447	AAPL	037833DD9	2047-09-12	SEN	AA	3.7500	13.9700	22.7800	79.2100	30.0000	22.4623	7.5377	1.7497	cheap	34,407,947.0000	1,000,000.0000
1452	AAPL	037833DQ0	2049-09-11	SEN	AA	2.9500	15.4000	24.8100	67.5900	31.0000	25.2360	5.7640	1.3380	cheap	97,827,660.0000	1,500,000.0000
1457	AAPL	037833EA4	2060-08-20	SEN	AA	2.5500	18.9400	35.9100	59.0600	25.0000	30.6141	-5.6141	-1.3032	rich	72,485,993.0000	1,750,000.0000
2368	AMZN	023135BU9	2060-06-03	SEN	AA	2.7000	17.9400	35.6900	58.0300	37.0000	26.6284	10.3716	2.5791	cheap	49,920,298.0000	2,000,000.0000
2372	AMZN	023135CC8	2061-05-12	SEN	AA	3.2500	17.1700	36.6400	65.8200	33.0000	25.1675	7.8325	1.9477	cheap	104,931,881.0000	1,750,000.0000
2361	AMZN	023135BC9	2027-08-22	SEN	AA	3.1500	2.2700	2.4300	97.7800	13.0000	6.8327	6.1673	1.5336	cheap	183,378,367.0000	3,500,000.0000
2378	AMZN	023135CJ3	2052-04-13	SEN	AA	3.9500	14.7700	27.4300	79.5000	17.0000	21.2109	-4.2109	-1.0471	rich	259,636,243.0000	2,500,000.0000
2362	AMZN	023135BF2	2037-08-22	SEN	AA	3.8750	9.5200	12.5800	89.8200	8.0000	13.8495	-5.8495	-1.4546	rich	122,483,763.0000	2,750,000.0000
2379	AMZN	023135CK0	2062-04-13	SEN	AA	4.1000	16.3800	37.5800	79.1300	14.0000	23.5253	-9.5253	-2.3687	rich	210,660,987.0000	1,250,000.0000
3647	BA	097023AE5	2031-09-15	SEN	BBB	8.7500	5.0500	6.5500	118.7600	69.0000	8.5157	60.4843	7.8562	cheap	10,636,728.0000	250,000.0000
3648	BA	097023AS4	2038-02-15	SEN	BBB	6.6250	8.6800	13.0700	106.6700	56.0000	16.4247	39.5753	5.1404	cheap	1,975,524.0000	300,000.0000
3660	BA	097023BZ7	2048-03-01	SEN	BBB	3.6250	13.4700	23.2500	67.0900	57.0000	27.1141	29.8859	3.8818	cheap	3,631,190.0000	350,000.0000
3652	BA	097023BK0	2035-03-01	SEN	BBB	3.3000	8.0900	10.0600	80.8500	49.0000	21.2426	27.7574	3.6054	cheap	4,287,303.0000	250,000.0000
3653	BA	097023BL8	2045-03-01	SEN	BBB	3.5000	12.7300	20.2100	68.4300	54.0000	26.4437	27.5563	3.5793	cheap	5,241,332.0000	250,000.0000
3650	BA	097023AX3	2039-03-15	SEN	BBB	6.8750	9.1100	14.1600	108.5500	39.0000	16.5238	22.4762	2.9194	cheap	23,494,774.0000	500,000.0000
4616	CAT	149123BL4	2035-09-15	SEN	A	5.3000	8.0100	10.6100	104.0400	32.0000	15.9897	16.0103	3.7149	cheap	6,383,130.0000	307,320.0000
4617	CAT	149123BN0	2036-08-15	SEN	A	6.0500	8.2800	11.5400	109.6500	27.0000	15.9174	11.0826	2.5715	cheap	15,123,063.0000	460,060.0000
4615	CAT	149123BJ9	2031-05-01	SEN	A	7.3000	4.8700	6.1700	115.2800	23.0000	11.9433	11.0567	2.5655	cheap	8,615,984.0000	242,226.0000
4621	CAT	149123CE9	2064-05-15	SEN	A	4.7500	15.9700	39.6900	86.9900	30.0000	19.8442	10.1558	2.3565	cheap	26,482,405.0000	500,000.0000
4624	CAT	149123CF6	2049-09-19	SEN	A	3.2500	14.9800	24.8300	70.0800	14.0000	20.1441	-6.1441	-1.4256	rich	42,413,979.0000	1,000,000.0000
4614	CAT	149123BE0	2097-03-01	SEN	A	7.3750	16.5700	72.9700	125.5900	3.0000	19.0046	-16.0046	-3.7136	rich	520,455.0000	246,115.0000
5865	DIS	254687EP7	2039-08-15	SEN	A	6.9000	9.3700	14.5800	114.3500	47.0000	23.8146	23.1854	4.0987	cheap	6,617,980.0000	236,418.0000
5850	DIS	25468P CR5	2041-12-01	SEN	A	4.1250	11.2800	16.9100	85.0700	41.0000	21.8700	19.1300	3.3818	cheap	41,489,553.0000	600,000.0000
5851	DIS	25468P CX2	2042-12-01	SEN	A	3.7000	11.9600	17.9300	79.4900	39.0000	22.0838	16.9162	2.9904	cheap	10,786,822.0000	500,000.0000
5874	DIS	254687FR2	2040-03-23	SEN	A	4.6250	10.5400	15.2000	93.9500	14.0000	21.6960	-7.6960	-1.3605	rich	66,663,925.0000	750,000.0000

	company_ symbol	CUSIP	MATU RITY	SECURI TY_LEVEL	RATING _CAT	COUP ON	DURA TION	TMT	PRICE EOM	T_Sprea d_bps	spread_fi t_bps	resid_ bps	resi d_z	rich_c heap	T_DVolum e_num	AMOUNT_OUT STANDING
58 55	DIS	25468P DN3	2046- 07-30	SEN	A	3.000 0	14.040 0	21.6 400	68.1600	7.0000	23.6981	-16.69 81	-2.95 19	rich	10,875,093 .0000	500,000.0000
58 63	DIS	254687 EF9	2037- 03-01	SEN	A	6.150 0	8.5100	12.0 900	108.470 0	4.0000	21.4539	-17.45 39	-3.08 55	rich	14,078,217 .0000	320,934.0000
75 21	T	00206R DH2	2042- 03-15	SEN	BBB	5.150 0	10.920 0	17.2 000	92.5100	67.0000	23.8863	43.113 7	5.81 56	cheap	6,674,022. 0000	316,928.0000
75 11	T	04650N AB0	2040- 09-01	SEN	BBB	5.350 0	10.280 0	15.6 500	97.1800	56.0000	23.3025	32.69 75	4.41 06	cheap	22,006,135 .0000	614,550.0160
75 39	T	00206R JF0	2041- 03-29	SEN	BBB	6.250 0	10.170 0	16.2 300	104.270 0	57.0000	25.6175	31.38 25	4.23 32	cheap	35,328,823 .0000	294,119.0000
75 19	T	00206R DF6	2040- 08-15	SEN	BBB	6.000 0	9.9300	15.6 000	101.620 0	51.0000	24.5017	26.49 83	3.57 44	cheap	8,902,949. 0000	400,455.0000
75 27	T	00206R DT6	2057- 03-01	SEN	BBB	5.700 0	14.300 0	32.3 800	96.7300	55.0000	31.8103	23.18 97	3.12 81	cheap	5,532,901. 0000	325,476.0000
75 15	T	00206R CG5	2044- 06-15	SEN	BBB	4.800 0	11.750 0	19.4 900	88.0800	45.0000	24.4279	20.57 21	2.77 50	cheap	12,746,445 .0000	455,205.0000

Rich/Cheap Signal on 2025-03-31



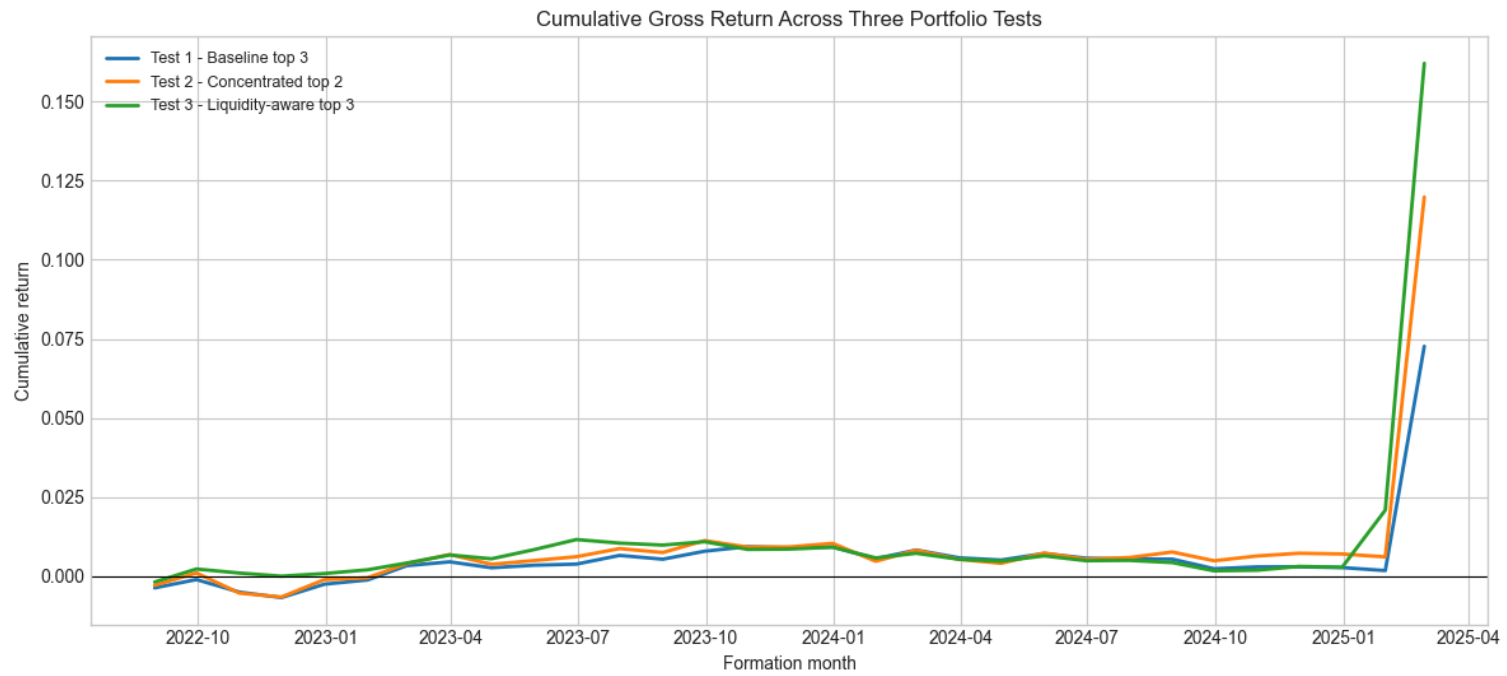
Three Duration-Neutral Portfolio Tests

Rather than choosing one construction and stopping, I run the backtest three ways:

1. **Baseline top 3:** select the three widest residual z-scores and the three tightest, with position size proportional to signal strength.
2. **Concentrated top 2:** use only the two strongest signals on each side. This increases concentration and may capture larger convergence moves, but it also raises idiosyncratic risk.
3. **Liquidity-aware top 3:** multiply the residual z-score by a capped square-root measure of TRACE dollar volume relative to the issuer-month median. This favors signals in bonds that appear easier to trade.

Every version scales the short side so $\text{sum}(\text{position} * \text{price} * \text{duration}) = 0$, then normalizes gross market value to one. Since $\text{DV01} = \text{price} * \text{duration} / 10,000$, this also makes first-order net DV01 approximately zero. It does not remove curve-shape, credit, downgrade, or liquidity risk.

trial	mont hs	gross_annualize d_return	gross_annualized _volatility	gross_sh arpe	gross_total return	win_r ate	average_tur nover	net_25bps_annualiz ed_return	net_25bps_ sharpe	net_25bps_tota l_return
Trial 1 - Baseline top 3	31	2.75%	4.47%	0.62	7.27%	51.6%	0.78	0.45%	0.10	1.16%
Trial 2 - Concentrated top 2	31	4.48%	7.10%	0.63	11.99%	58.1%	0.82	2.02%	0.29	5.31%
Trial 3 - Liquidity- aware top 3	31	5.99%	8.66%	0.69	16.21%	54.8%	0.79	3.60%	0.42	9.56%



Selected portfolio: Trial 3 - Liquidity-aware top 3

sample	annualized_return	annualized_volatility	annualized_sharpe	total_return
All six issuers	5.99%	8.66%	0.69	16.21%
Excluding AAPL	0.05%	0.77%	0.06	0.12%

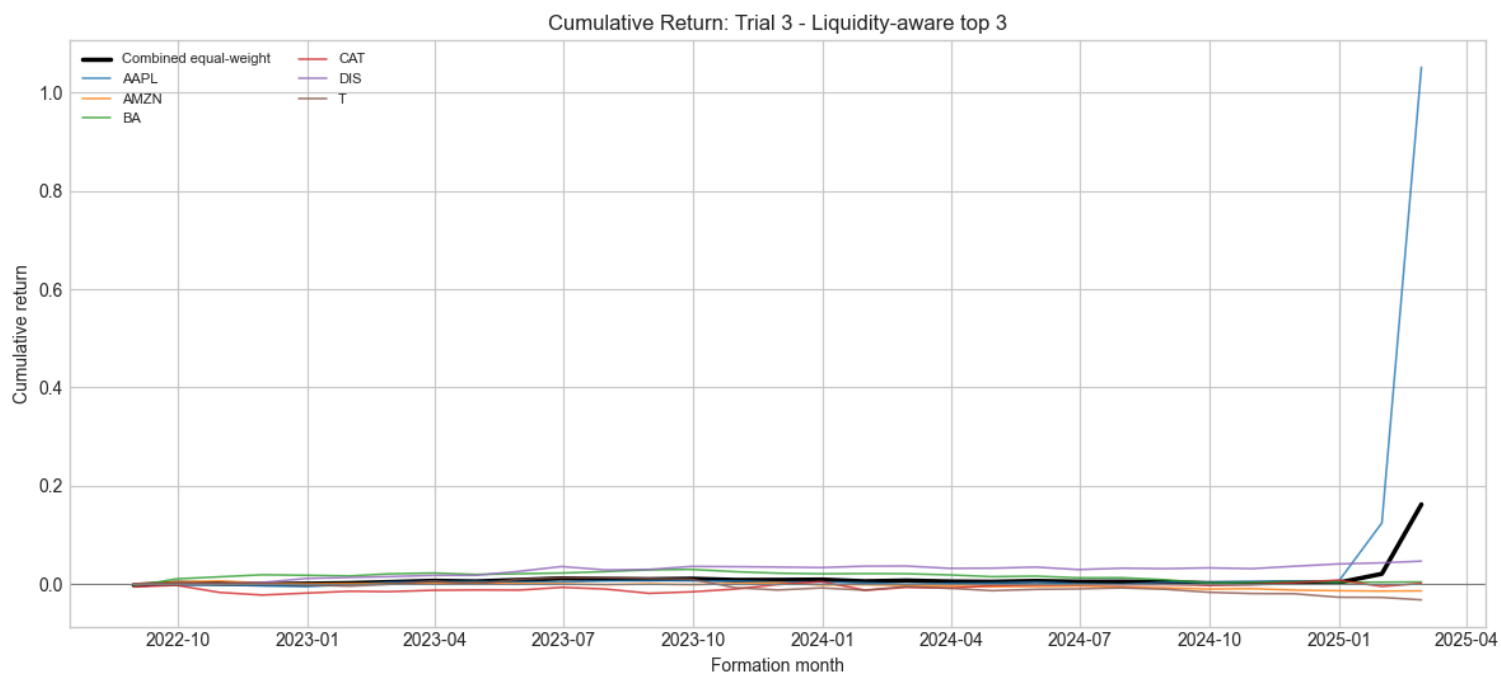
Results for the Selected Portfolio

I select the version with the highest Sharpe ratio after a 25 bp one-way turnover charge. This rule chooses the liquidity-aware top-three portfolio. The table above also reports the other two attempts, so the final result can be compared with the original baseline.

The selection is in-sample. It is useful for understanding how portfolio construction changes the result, but it is not an unbiased estimate of future performance. The AAPL exclusion check helps show whether the result is broad or driven by one issuer.

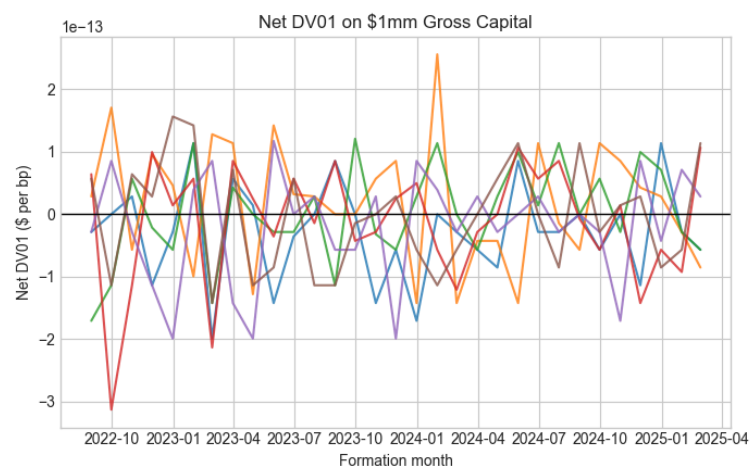
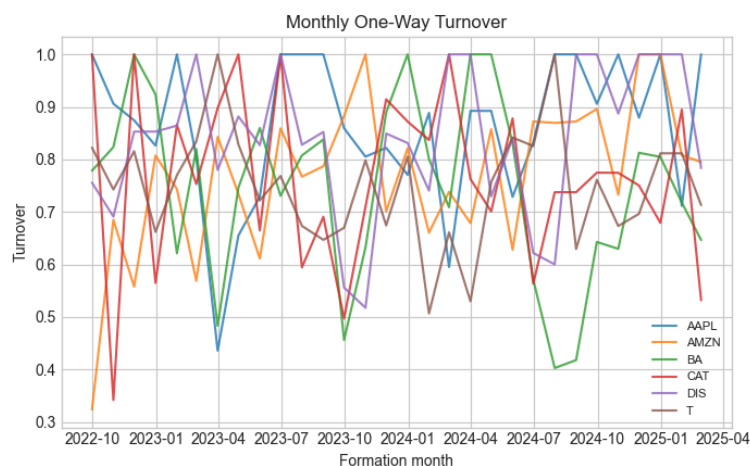
Dollar PnL below assumes \$1,000,000 of gross capital. The combined portfolio equal-weights the available issuer portfolios each month.

	month s	mean_monthl y_return	monthly_v olatility	annualized _return	annualized_ volatility	annualized _sharpe	total_re turn	win_r ate	total_pn l_usd	avg_tur nover	avg_abs_portfoli o_duration	avg_abs_net_ dv01_usd	avg_pos itions
AAPL	31.000 000	3.06%	14.90%	32.08%	51.62%	0.62	105.19 %	54.8 %	\$948,89 5	0.86	0.000000	\$0.0000	6.0
AMZN	31.000 000	-0.04%	0.19%	-0.54%	0.67%	-0.81	-1.38%	45.2 %	\$-13,848	0.77	0.000000	\$0.0000	6.0
BA	31.000 000	0.01%	0.39%	0.17%	1.36%	0.13	0.44%	54.8 %	\$4,644	0.75	0.000000	\$0.0000	6.0
CAT	31.000 000	0.01%	0.64%	0.06%	2.23%	0.03	0.15%	61.3 %	\$2,129	0.77	0.000000	\$0.0000	6.0
DIS	31.000 000	0.15%	0.36%	1.78%	1.25%	1.42	4.67%	67.7 %	\$45,861	0.84	0.000000	\$0.0000	6.0
T	31.000 000	-0.10%	0.51%	-1.26%	1.78%	-0.71	-3.21%	35.5 %	\$-32,242	0.75	0.000000	\$0.0000	6.0
combi ned	31.000 000	0.51%	2.50%	5.99%	8.66%	0.69	16.21%	54.8 %	\$159,24 0	0.79	0.000000	\$0.0000	36.0
	annualized_return			annualized_volatility			annualized_sharpe			total_return		win_rate	
0 bps	5.99%			8.66%			0.69			16.21%		54.8%	
10 bps	5.03%			8.66%			0.58			13.51%		41.9%	
25 bps	3.60%			8.66%			0.42			9.56%		22.6%	
50 bps	1.26%			8.67%			0.15			3.28%		9.7%	



Turnover and DV01 Neutrality Checks

The first-order hedge is verified in both duration-contribution units and dollars of DV01 for \$1,000,000 gross capital. Values near zero are expected up to floating-point precision. Gross DV01 is nonzero; the purpose of the long-short scaling is to make the signed net DV01 approximately zero.



company_symbol	max_abs_net_dv01_usd	avg_abs_net_dv01_usd	avg_gross_dv01_usd	avg_turnover
AAPL	\$0.000000	\$0.000000	\$1,426	0.86
AMZN	\$0.000000	\$0.000000	\$1,339	0.77
BA	\$0.000000	\$0.000000	\$981	0.75
CAT	\$0.000000	\$0.000000	\$998	0.77
DIS	\$0.000000	\$0.000000	\$1,133	0.84
T	\$0.000000	\$0.000000	\$1,058	0.75

Outlier Check

Professor Boroditsky discussed several reasons why a bond can look mispriced without offering a clean convergence trade: low liquidity, small issue size, age, coupon structure, optionality, seniority, and issuer-entity differences. The first table identifies bonds that appear repeatedly in the selected portfolio. The second compares those observations with bonds that were not selected.

This comparison is descriptive. If the cheap bonds are consistently smaller or less liquid, part of the residual may be compensation for trading difficulty rather than free alpha.

	company_ symbol	CUSIP	BOND TYPE	MATU RITY	COUP ON	times_sel ected	times_c heap	times_ rich	avg_dur ation	avg_sprea d_bps	avg_abs_re sid_bps	median_dv olume	median_amount_ outstanding	avg_bond_a ge_years
51	AMZN	023135 CK0	CDEB	2062- 04-13	4.100 0	21	12	9	17.2395	39.8095	12.2362	78,112,968. 0000	1,250,000.0000	1.5984
83	BA	097023 CW3	CDEB	2050- 05-01	5.805 0	21	0	21	12.8257	20.8571	11.8276	783,854,16 6.0000	5,500,000.0000	3.5372
50	AMZN	023135 CJ3	CDEB	2052- 04-13	3.950 0	21	5	16	15.7210	29.2857	7.8069	235,968,72 7.0000	2,500,000.0000	1.5310
19 6	T	00206R MT6	CDEB	2034- 02-15	5.400 0	20	0	20	7.4315	12.8000	12.3884	830,164,16 5.5000	2,750,000.0000	0.9132
95	CAT	149123 BN0	CDEB	2036- 08-15	6.050 0	19	16	3	8.8621	44.6316	24.0086	15,590,008 .0000	460,060.0000	17.3524
10 2	CAT	149123 CJ8	CDEB	2050- 04-09	3.250 0	17	9	8	15.9841	37.8235	11.3302	44,915,056 .0000	1,200,000.0000	3.3510
94	CAT	149123 BL4	CDEB	2035- 09-15	5.300 0	16	15	1	8.5388	43.1875	23.6879	9,160,466. 0000	307,320.0000	18.5878
98	CAT	149123 CD1	CDEB	2044- 05-15	4.300 0	15	8	7	12.9540	40.0000	30.0960	12,269,473 .0000	500,000.0000	9.4668
14 8	DIS	254687 FS0	CDEB	2050- 03-23	4.700 0	15	5	10	14.4193	31.2667	13.4823	84,354,542 .0000	1,750,000.0000	3.4940
37	AMZN	023135 BJ4	CDEB	2047- 08-22	4.050 0	15	10	5	14.3487	37.5333	10.2495	121,313,48 7.0000	3,487,493.0000	5.3109
56	BA	097023 AE5	CDEB	2031- 09-15	8.750 0	14	14	0	5.5793	67.5000	50.5830	16,518,946 .5000	250,000.0000	32.4488
10 0	CAT	149123 CF6	CDEB	2049- 09-19	3.250 0	14	2	12	15.5221	25.0000	15.3317	25,497,276 .5000	1,000,000.0000	4.3313
45	AMZN	023135 CA2	CDEB	2041- 05-12	2.875 0	14	11	3	12.8543	40.2857	14.5116	77,543,957 .5000	2,000,000.0000	2.3133
47	AMZN	023135 CC8	CDEB	2061- 05-12	3.250 0	14	7	7	17.9786	37.8571	13.8536	48,898,063 .5000	1,750,000.0000	2.6651
27	AAPL	037833 EQ9	CDEB	2052- 08-08	3.950 0	14	4	10	15.8200	33.4286	13.1493	283,096,02 7.5000	1,750,000.0000	1.1168
38	AMZN	023135 BM7	CDEB	2057- 08-22	4.250 0	14	5	9	16.4143	30.5714	9.7887	72,689,683 .0000	2,247,465.0000	5.9722
99	CAT	149123 CE9	CDEB	2064- 05-15	4.750 0	13	5	8	16.7354	35.4615	23.5855	8,786,039. 0000	500,000.0000	9.6617
23	AAPL	037833 EK2	CDEB	2051- 08-05	2.700 0	13	2	11	16.9262	26.7692	14.1643	62,511,350. 0000	1,800,000.0000	2.3322
28	AAPL	037833 ER7	CDEB	2062- 08-08	4.100 0	13	4	9	17.1800	32.1538	11.3181	133,663,34 6.0000	1,250,000.0000	1.4791
46	AMZN	023135 CB0	CDEB	2051- 05-12	3.100 0	13	4	9	16.0708	27.2308	10.9342	102,776,74 8.0000	3,250,000.0000	2.7252

	observatio ns	median_abs_residual_ bps	median_dollar_vol ume	median_amount_outstan ding	median_bond_age_y ears	median_durati on	median_t mt	median_coup on
selection_gro up								
long cheap	558	22.3119	35,557,574.0000	750,000.0000	5.4552	12.3200	18.9550	4.2000
not selected	6443	3.9493	45,349,053.0000	1,000,000.0000	4.4162	6.8600	8.2400	3.6250
short rich	558	10.3933	109,797,947.0000	1,750,000.0000	3.4442	13.2100	23.7500	4.1000

What I Found

The final cell turns the main tables into a short written conclusion. I keep the three-test comparison, transaction costs, issuer concentration, and data limitations in the discussion so the result is not presented more strongly than the evidence supports.

Summary Write-up

I analyzed AAPL, AMZN, BA, CAT, DIS, and T from 2022-08-31 through 2025-03-31. For each issuer-month, I kept a common seniority and rating category. I also excluded defaulted and convertible bonds, required at least 1 year to maturity, and required about \$200 million outstanding.

I fitted each issuer curve with log-duration, duration, and coupon. TRACE dollar volume and amount outstanding receive more weight because larger, actively traded issues generally give cleaner pricing observations. A positive fitting residual is treated as cheap and a negative residual as rich. The long-cheap observations have lower median dollar volume than the non-selected observations, so liquidity still appears to be part of the signal.

The three portfolio tests produced noticeably different results. The baseline top-three version earned 2.75% annualized before costs. Concentrating in the top two signals raised that figure to 4.48%. The liquidity-aware top-three version earned 5.99% and had the strongest 25 bp cost-adjusted Sharpe ratio, so I use it as the final specification. This choice was made on the same data used for evaluation, which means the reported winner is subject to selection bias.

For the selected portfolio, gross total return is 16.21%, annualized return is 5.99%, annualized volatility is 8.66%, and annualized Sharpe ratio is 0.69. The average one-way monthly turnover is 0.79. With a 25 bp one-way cost, annualized return falls to 3.60% and Sharpe falls to 0.42. On the \$1 million gross-capital convention, the sum of monthly gross PnL is \$159,240.

The DV01 hedge works mechanically: average absolute net DV01 is \$0.000000 per basis point. The return is less broad than the headline number suggests. When AAPL is removed, annualized return falls to 0.05% and Sharpe falls to 0.06. Among the six issuers, DIS has the strongest result and AMZN the weakest.

95.6% of filtered bond-month observations have a valid next-month WRDS `RET_EOM`. The price-plus-accrual fallback was available but was not needed in this sample. I view the backtest as evidence that construction choices matter, not as a fully tradable strategy. Bid-offer spreads, financing, borrow availability, fill probability, call features, and point-in-time data delays are not completely modeled.

Main Assumptions and Limitations

- **Return measure:** I use WRDS RET_EOM as the monthly total return. The fallback adds clean-price return and accrued coupon using actual gap days over 360, but it does not rebuild the full coupon schedule.
- **Timing:** I assume the month-end fields were available when the position was formed. Publication delays and later revisions are not modeled.
- **Curve model:** the three-factor regression is transparent and matches the assignment, but it is not a full Nelson-Siegel cash-flow model and does not rebuild the Treasury zero curve.
- **Bond features:** I control for seniority, rating, default status, convertibility, maturity, and issue size. Complete call schedules, 144A status, subsidiary distinctions, insurance, and step-up coupons are not available in the supplied files.
- **Risk hedge:** the portfolio neutralizes first-order parallel-rate DV01. It can still carry Treasury-curve slope and curvature risk, spread-duration risk, issuer beta, downgrade risk, and liquidity exposure.
- **Trading costs:** the turnover-based cost cases are approximations. Actual bid-offer spreads, market impact, financing, borrow fees, and failed fills may be higher and may differ between buys and shorts.
- **Model selection:** I tested three portfolio rules and selected the best one on this sample. That improves the in-sample result but introduces selection bias. A cleaner test would choose the rule on an earlier training period and evaluate it on later data.
- **Issuer concentration:** the robustness check shows that AAPL contributes a large share of the selected portfolio's return.
- **Shorting:** some corporate bonds are expensive or impossible to borrow. The long-short backtest is a research exercise, not proof that every trade could have been executed.